CHITRANG POTDAR

Backend Engineer (Python) · FastAPI | AWS | Docker · AI/LLM Systems

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PROFESSIONAL SUMMARY

Backend Engineer with production experience designing and shipping Python-based systems end-to-end. Sole backend engineer on an Investment Operations Platform — owning APIs, CI/CD pipelines, Redis-based caching, Docker containerisation, and AWS deployment. Additional experience architecting serverless AWS pipelines and integrating AI/LLM capabilities into backend workflows, including audio transcription and structured output generation using Claude API.

TECHNICAL SKILLS

Backend	Python, Flask, FastAPI, RESTful API Design, JWT Authentication, 2FA (TOTP), Redis
Databases	PostgreSQL, Microsoft SQL Server — schema design, indexing, query optimisation
DevOps & Cloud	Docker, GitHub Actions, CI/CD Pipelines, AWS EC2, AWS Lambda, AWS S3, AWS Transcribe, AWS API Gateway
Testing	Pytest, Unit Testing, Coverage Reporting, Playwright (test automation)
AI/ML	Claude API, CrewAI, MLflow, ZenML, Scikit-learn, XGBoost, NeuralProphet, Pandas, NumPy
Languages	Python (primary), SQL, JavaScript
Tools	Git, GitHub, Postman, Linux

EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune*Oct 2024 – Present*

Investment Operations Platform

Stack: *Python, Flask, PostgreSQL, Redis, Docker, GitHub Actions, AWS EC2, Pytest, Playwright*

- Sole backend engineer for an Investment Operations Platform — owning the full backend lifecycle across feature development, debugging, testing, and production releases.
- Designed and implemented REST APIs for user management, transaction handling, and financial reporting — enforcing structured input validation, error handling, and consistent response contracts across all endpoints.
- Designed backend services with clear separation of concerns and structured validation to improve maintainability and reduce debugging overhead.
- Owned and implemented CI/CD pipelines and containerisation using GitHub Actions and Docker, ensuring consistent and reliable deployment workflows while eliminating manual release processes.
- Engineered Redis-based session management — auth tokens cached on login and served from Redis, eliminating redundant database queries on high-frequency authentication endpoints.
- Achieved 80%+ unit test coverage using Pytest, reducing regression defects across core API modules.
- Automated internal workflows using Playwright, saving ~2–3 hours/week of manual operational effort.

Serverless Data Capture Pipeline

Stack: *AWS Lambda, AWS API Gateway, Google Sheets API*

- Designed and implemented a serverless data capture pipeline using AWS Lambda and API Gateway, routing frontend submissions into Google Sheets and eliminating the need for a dedicated database in this workflow.

AI-Powered Meeting Intelligence Pipeline

Stack: *AWS Lambda, AWS API Gateway, AWS S3, AWS Transcribe, Claude (Anthropic API)*

- Architected an end-to-end serverless backend pipeline for meeting analysis — audio recordings and agenda documents are ingested from S3, transcribed via AWS Transcribe, and the resulting transcript is passed to Claude API alongside the agenda for structured analysis.
- Engineered the Claude API integration to produce structured outputs: a concise meeting summary, agenda adherence scoring (on-agenda vs. off-agenda coverage classification), and an auto-generated actionable task list — returned to the client via API Gateway.

Python Developer Intern · ArrayPointer Pvt. Ltd., Pune*Dec 2023 – Apr 2024*

Financial Document Parsing Pipeline

Stack: *Python, PyPDF2, Camelot, Tabula, Pandas*

- Designed and developed an automated PDF parsing pipeline for financial documents (electricity bills, CAS statements), reducing manual extraction effort by ~70% and delivering clean, structured JSON output.

- Engineered a multi-strategy parsing layer combining PyPDF2, Camelot, and Tabula to handle diverse and inconsistent document layouts with consistent accuracy.

Revenue Forecasting Model
Stack: *Python, XGBoost, Pandas, Scikit-learn*

- Developed a time-series revenue forecasting model using XGBoost with incremental learning — forecast accuracy improved by ~20% as new data was ingested, designed for continuous production use without full retraining.

PROJECTS

TaskHub — Project Management Platform *Aug – Sep 2025*
GitHub: github.com/Dev-Chitrang/TaskHub-Full-Stack-Project
Stack: *FastAPI, PostgreSQL, Docker*

- Designed and developed a scalable backend system using FastAPI and PostgreSQL supporting authentication, RBAC, workspace management, and task operations.
- Implemented JWT authentication with email verification, 2FA (TOTP), and password recovery; containerised with Docker using dev/prod config separation and structured API logging.

Retail DataOps & MLOps Platform *Mar – Jul 2025*
GitHub: github.com/Dev-Chitrang/data-warehouse-to-mlops
Stack: *FastAPI, MSSQL, Scikit-learn, ZenML, MLflow, NeuralProphet, Docker*

- Engineered ETL pipelines ingesting ERP/CRM data into a structured SQL Server data warehouse (Bronze/Silver/Gold layers); exposed Demand Forecasting, Customer Segmentation, and Product Recommendation model outputs via FastAPI endpoints. Full stack containerised with Docker.
- Managed ML experiment tracking with MLflow and orchestrated model training and deployment pipelines with ZenML for reproducibility.

AI Trip Planner — Multi-Agent System *Sep – Oct 2025*
GitHub: github.com/Dev-Chitrang/Multi-Agent-Trip-Planner
Stack: *CrewAI, Gemini 2.0 Flash, SerpAPI, Python*

- Designed and developed a multi-agent AI system using CrewAI — coordinating specialised Flight, Hotel, and Itinerary Planner agents powered by Gemini 2.0 Flash, with real-time data retrieval via SerpAPI.

EDUCATION

B.E. Computer Engineering · Savitribai Phule Pune University, Pune *CGPA: 8.56 / 10 | Jun 2025*

RESEARCH & ACHIEVEMENTS

- Published: Corporate Credit Rating Analysis using AI & Sentiment Analysis — IJIRCCE (Impact Factor: 8.65), Nov 2024.
- Research repository: github.com/Dev-Chitrang/Final-year-project